

A Six-Parameter Extension of the Direct Somos–4 Algebraic Generating Function

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Abstract

Our earlier work constructed a five-parameter algebraic generating function whose Hankel determinants start with (λ, m, r, η) and obey a Somos–4 recurrence. The natural next step is to free the coefficient of the first product term:

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2.$$

This adds one more parameter, giving us six total. The construction itself is straightforward: set $q^6 = \alpha$ and rescale the previous five parameters by suitable powers of q , then replace x with qx . The result is a Hankel sequence with the desired six parameters, while the first four determinants remain exactly λ, m, r, η .

But here a subtle point arises. The determinant sequence itself is rational in the six parameters, as is the quartic curve it generates. Yet our transported generating function is not: the coefficient of x becomes $q^{-5}r$, involving a sixth root of α . We can thus write the determinants and curve over the rational field without adjoining any root, yet this particular power-series lift requires one.

A comparison with Chang and Liu’s quadratic-orthogonal-pair construction illuminates these phenomena. With a natural shift, their bilateral Somos–4 sequence becomes ours, and their subdiagonal data satisfy $d_n = B_{n+2}$. Their square-root choice $u^2 = \alpha$ also admits a clean interpretation: replacing u by $-u$ corresponds exactly to replacing a generating series $F(x)$ by $F(-x)$. This operation fixes every ordinary Hankel determinant but multiplies the once-shifted determinant of order N by $(-1)^N$. So ordinary Hankel determinants lose a parity choice that the shifted ones preserve.

1 Introduction

The preceding paper [3] started with four prescribed Hankel determinants,

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta),$$

and constructed an algebraic generating function satisfying

$$H_{k+4}H_k = H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{1}$$

Here we relax that construction by allowing the leading coefficient to vary:

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{2}$$

We thus gain six independent parameters in place of five: $(\lambda, m, r, \eta, \alpha, \tau)$, with the special case $\alpha = 1$ recovering the earlier construction.

Two questions present themselves, distinct in both scope and difficulty.

The existence question is the simpler one: can we extend the old construction so that the first four Hankel determinants remain exactly (λ, m, r, η) and the recurrence becomes (2)? The answer is yes, and Section 2 provides an explicit weighted rescaling.

The *rationality question* is harder. Can we write such a quadratic algebraic generating function using only rational functions of the six parameters over the field

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau),$$

while preserving the shifted-determinant identity $D_N = H_{N+2}$? Our weighted construction does not achieve this: it introduces a sixth root q of α , and already its coefficient of x is $q^{-5}r$. This is not a cosmetic radical that simplifies away.

We need a few preliminary terms. By an *ordinary Hankel sequence* we mean the determinants H_N of a coefficient sequence. A *once-shifted determinant* D_N uses the same coefficients but with every index increased by one. We call a coefficient sequence a *moment sequence*, adopting the language of continued fractions and orthogonal polynomials, though here “moments” are simply the coefficients of a formal power series—no measure is involved. The *Somos-4 orbit* is the sequence obtained by iterating the recurrence for fixed initial values.

A key observation throughout this work is that three related objects carry different information:

Hankel determinant sequence, algebraic curve, generating series.

The first two are rational in the six parameters. The third need not be. We will write “descends to K ” as shorthand for “can be written with coefficients in K ”. By this measure, the determinant sequence and the curve descend, while our particular transported series does not.

This distinction clarifies our position relative to recent work. Liu, Wang, and Zhang [2] give broad rational-function constructions for generic (α, β) -Somos-4 Hankel transforms. Our problem is narrower: we require four initial determinants to be independently prescribed, and we insist on a specific shifted-determinant normalization. So the scope is more restricted, though the geometric picture is richer.

Here is the plan. Section 2 performs the weighted rescaling and proves the six-parameter recurrence. Section 3 tracks the shifted determinants, showing exactly where fractional powers of α first appear. Section 4 establishes that the determinant sequence and curve both descend to K . Section 5 translates our construction into Chang and Liu’s notation and identifies the Jacobi subdiagonal data. Section 6 interprets the sign choice as parity conjugation. We conclude with a summary of the main facts.

2 From the five-parameter construction to six parameters

2.1 The five-parameter input

Let

$$G_{1;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} g_k x^k$$

be the algebraic series from [3]. For any coefficient sequence we define the ordinary and once-shifted Hankel determinants by

$$H_N(g) = \det(g_{i+j})_{0 \leq i,j \leq N-1}, \quad D_N(g) = \det(g_{i+j+1})_{0 \leq i,j \leq N-1}.$$

That construction satisfies

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta), \quad D_N = H_{N+2}. \tag{3}$$

The underlying curve has discriminant

$$\Delta_1(x) = (x\lambda^2\eta mr^2)^2 \left((1 - Jx + \tau x^2)^2 - 4x^3 \right), \tag{4}$$

where

$$J = \frac{\lambda\eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda\eta}. \tag{5}$$

For our purposes, these two facts suffice.

2.2 Weighted transport

The key idea is to introduce a sixth root of α and use it to rescale carefully. Set

$$\mathbb{K} = \mathbb{Q}(\lambda, m, r, \eta, q, \tau), \quad q^6 = \alpha,$$

and define new parameters

$$\lambda' = \lambda, \quad m' = q^{-2}m, \quad r' = q^{-6}r, \quad \eta' = q^{-12}\eta, \quad \tau' = q^{-8}\tau. \quad (6)$$

Now form the transported series

$$G_{\alpha; \lambda, m, r, \eta, \tau}(x) := G_{1; \lambda', m', r', \eta', \tau'}(qx). \quad (7)$$

If the original has expansion $G_{1; \lambda', m', r', \eta', \tau'}(x) = \sum_{k \geq 0} a_k x^k$ and the transported has $G_{\alpha; \lambda, m, r, \eta, \tau}(x) = \sum_{k \geq 0} b_k x^k$, then

$$b_k = q^k a_k. \quad (8)$$

Proposition 1 (Determinant weights). *For every $N \geq 1$,*

$$H_N(b) = q^{N(N-1)} H_N(a), \quad D_N(b) = q^{N^2} D_N(a). \quad (9)$$

Proof. In the ordinary Hankel matrix, its (i, j) entry is multiplied by q^{i+j} . Factoring q^i from row i and q^j from column j gives the exponent $2 \sum_{j=0}^{N-1} j = N(N-1)$. For the shifted matrix the entry is multiplied by q^{i+j+1} . The additional factor q in every row contributes N , so the total exponent is $N(N-1) + N = N^2$. $\square$

2.3 The six-parameter Somos-4 recurrence

Theorem 2. *Let $\hat{H}_N = H_N(b)$. Then*

$$(\hat{H}_1, \hat{H}_2, \hat{H}_3, \hat{H}_4) = (\lambda, m, r, \eta)$$

and

$$\hat{H}_{k+4} \hat{H}_k = \alpha \hat{H}_{k+3} \hat{H}_{k+1} + \tau \hat{H}_{k+2}^2, \quad k \geq 1. \quad (10)$$

Proof. Equation (9) and (6) give

$$\hat{H}_1 = \lambda, \quad \hat{H}_2 = q^2 m' = m, \quad \hat{H}_3 = q^6 r' = r, \quad \hat{H}_4 = q^{12} \eta' = \eta.$$

Write $H'_N = H_N(a)$. The primed sequence satisfies

$$H'_{k+4} H'_k = H'_{k+3} H'_{k+1} + \tau' H_{k+2}'^2.$$

The weight difference between the left-hand product and the first product on the right is 6, while that between the left-hand product and $H_{k+2}'^2$ is 8. Hence

$$\hat{H}_{k+4} \hat{H}_k = q^6 \hat{H}_{k+3} \hat{H}_{k+1} + q^8 \tau' \hat{H}_{k+2}^2.$$

Now use $q^6 = \alpha$ and $q^8 \tau' = \tau$. $\square$

Corollary 3 (Low-order recurrence check). *The first two values forced beyond the initial tuple are*

$$\hat{H}_5 = \frac{\alpha \eta m + \tau r^2}{\lambda}, \quad \hat{H}_6 = \frac{\alpha \hat{H}_5 r + \tau \eta^2}{m}.$$

These formulas provide a direct indexing check for any coefficient expansion of (7).

3 Why fractional powers appear

The weighted rescaling delivers the correct ordinary Hankel determinants, but the shifted determinants behave differently. Understanding this gap is the key to the rationality problem.

3.1 The transported companion identity

Proposition 4. *If the primed series obeys $D_N(a) = H'_{N+2}$, then*

$$\widehat{D}_N := D_N(b) = q^{-3N-2} \widehat{H}_{N+2}, \quad N \geq 1. \quad (11)$$

Proof. By (9),

$$D_N(b) = q^{N^2} H'_{N+2}, \quad \widehat{H}_{N+2} = q^{(N+2)(N+1)} H'_{N+2}.$$

Their exponent difference is $N^2 - (N+2)(N+1) = -3N - 2$. $\square$

Remark 5. A constant coefficient rescaling $b_k \mapsto cb_k$ changes the ratio $D_N(b)/\widehat{H}_{N+2}$ by c^{-2} . It cannot remove the factor q^{-3N} . Thus the literal identity $D_N = H_{N+2}$ is not invariant under this graded transport. Equation (11), rather than the ungraded identity, is the companion normalization naturally inherited from the five-parameter series.

3.2 Parity conjugation and the companion orientation

For any formal series

$$F(x) = \sum_{n \geq 0} f_n x^n, \quad P(F)(x) := F(-x) = \sum_{n \geq 0} (-1)^n f_n x^n,$$

the involution P has different effects on the ordinary and once-shifted Hankel determinants.

Proposition 6 (Parity determinant laws). *For every $N \geq 1$,*

$$H_N(P(F)) = H_N(F), \quad D_N(P(F)) = (-1)^N D_N(F). \quad (12)$$

Proof. The (i, j) entry of the ordinary Hankel matrix is multiplied by $(-1)^{i+j}$. Factoring $(-1)^i$ from row i and $(-1)^j$ from column j gives the total exponent $2 \sum_{j=0}^{N-1} j = N(N-1)$, which is even. In the once-shifted matrix the entry is multiplied by $(-1)^{i+j+1}$. The additional factor -1 in every row contributes $(-1)^N$, proving the second identity. $\square$

Corollary 7 (Parity-conjugate companion law). *If $D_N(F) = c_N H_{N+2}(F)$, then*

$$D_N(P(F)) = (-1)^N c_N H_{N+2}(P(F)). \quad (13)$$

In particular, the ordinary Hankel orbit cannot distinguish $F(x)$ from $F(-x)$, whereas a prescribed companion law distinguishes the two parity orientations whenever some odd-order companion determinant is nonzero.

Applied to our transported series, parity conjugation preserves the six-parameter Somos-4 orbit yet alters the graded companion identity:

$$D_N(G_\alpha(-x)) = (-1)^N q^{-3N-2} H_{N+2}(G_\alpha(-x)). \quad (14)$$

The upshot is concrete: $F(x)$ and $F(-x)$ share the same ordinary Hankel determinants but differ by a predictable sign in their shifted determinants. We reserve the term “orientation” for this sign choice alone. No interpretation via algebraic sheets or bilateral Somos reversal is required.

3.3 The first coefficients and failure of descent

The companion normalization of the five-parameter series gives $g_0 = \lambda$ and $g_1 = r$. When we apply the rescaling (6), we get $a_0 = \lambda$ and $a_1 = r' = q^{-6}r$. By (8),

$$b_0 = \lambda, \quad b_1 = q^{-5}r = \frac{qr}{\alpha}. \quad (15)$$

Expanding further:

$$b_2 = \frac{m}{\lambda} + \frac{q^{-10}r^2}{\lambda} = \frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda}, \quad (16)$$

$$b_3 = \frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2}. \quad (17)$$

Putting these together:

$$\begin{aligned} G_\alpha(x) = & \lambda + \frac{qr}{\alpha}x + \left(\frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda} \right) x^2 \\ & + \left(\frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2} \right) x^3 + O(x^4). \end{aligned} \quad (18)$$

The coefficient b_1 also emerges from (11) at $N = 1$:

$$b_1 = D_1(b) = q^{-5}H_3(b) = q^{-5}r.$$

So the fractional power is not an artifact of poor simplification; it is forced directly by the shifted-determinant relation at the first step.

Put

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau), \quad L = K(q), \quad q^6 = \alpha.$$

Proposition 8 (Non-descent of the transported branch). *In the generic case $r \neq 0$, the transported series (7) does not belong to $K[[x]]$.*

Proof. The coefficient $b_1 = q^{-5}r = qr/\alpha$ lies outside K generically. By Eisenstein's criterion, the polynomial $X^6 - \alpha$ is irreducible over K , so $q \notin K$. Since $r \neq 0$, we have $b_1 \notin K$, and thus the series does not lie in $K[[x]]$. $\square$

The radicals do not descend to K , yet they vanish from the ordinary Hankel determinants. For instance,

$$H_2(b) = b_0b_2 - b_1^2 = \lambda \left(\frac{m}{\lambda} + \frac{q^{-10}r^2}{\lambda} \right) - q^{-10}r^2 = m.$$

This example crystallizes the distinction: fractional powers appear in the coefficients but cancel in the determinants.

4 What is defined over the six-parameter field

The non-descent we proved applies to one particular generating series. It says nothing about the recurrence itself or its algebraic curve. Let us check each separately.

4.1 Transport of the discriminant

To track how the discriminant transforms, decompose the invariant J as

$$A = \frac{\lambda\eta}{mr} + \frac{\tau mr}{\lambda\eta}, \quad B = \frac{m^2}{\lambda r} + \frac{r^2}{m\eta}. \quad (19)$$

Under the rescaling (6),

$$J' = q^{-4}A + q^2B. \quad (20)$$

The substitution $x \mapsto qx$ then gives

$$qJ' = q^{-3}A + q^3B = \alpha^{-1/2}A + \alpha^{1/2}B. \quad (21)$$

Proposition 9 (Transported discriminant). *The discriminant obtained by substituting (6) and $x \mapsto qx$ into (4) is*

$$\Delta_\alpha(x) = q^{-50}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (q^{-3}A + q^3B)x + q^{-6}\tau x^2 \right)^2 - 4q^3x^3 \right]. \quad (22)$$

Equivalently,

$$\Delta_\alpha(x) = \alpha^{-25/3}(x\lambda^2\eta mr^2)^2 \left[\left(1 - (\alpha^{-1/2}A + \alpha^{1/2}B)x + \frac{\tau}{\alpha}x^2 \right)^2 - 4\alpha^{1/2}x^3 \right]. \quad (23)$$

Proof. The square prefactor has total q -weight $2(1 - 12 - 2 - 12) = -50$. The three terms in the residual factor transform as $J'x \mapsto qJ'x$, $\tau'x^2 \mapsto q^2\tau'x^2 = q^{-6}\tau x^2$, and $x^3 \mapsto q^3x^3$. Equations (20) and $q^6 = \alpha$ give the stated forms. $\square$

The overall monomial factor in (22) is a square in $\mathbb{K}(x)$ and does not affect the residual double cover. It should, however, be retained when comparing discriminants of particular polynomial representatives, because multiplying a quadratic equation by a scalar squares that scalar in its discriminant.

4.2 A rational model of the curve

Dropping the square prefactor from (23) and setting

$$x = \alpha^{1/2}z, \quad (24)$$

the residual part becomes

$$Y^2 = \left(1 - (A + \alpha B)z + \tau z^2 \right)^2 - 4\alpha^2 z^3. \quad (25)$$

Every coefficient here lies in $\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$.

To match the quartic convention from our earlier work [3], let $X = z^{-1}$ and $y = z^{-2}Y$, or equivalently $Y = z^2y$. Then (25) is birational to

$$y^2 = \left(X^2 - (A + \alpha B)X + \tau \right)^2 - 4\alpha^2 X. \quad (26)$$

Thus the curve descends to the six-parameter field. However, the change of variables (24) rescales power-series coefficients and destroys the Hankel normalization. So removing radicals from the curve is not the same as removing them from a specific generating series.

4.3 Formal representatives with the same Hankel determinants

Hankel determinants do not uniquely specify a generating series. For a nondegenerate sequence, define

$$\beta_n = \frac{H_{n-1}H_{n+1}}{H_n^2}, \quad H_0 = 1. \quad (27)$$

All H_n from our recurrence, and hence all β_n , lie in K . For any choice of $a_0, a_1, \dots \in K$, the Jacobi continued fraction

$$\tilde{G}(x) = \frac{H_1}{1 - a_0x - \frac{\beta_1x^2}{1 - a_1x - \frac{\beta_2x^2}{1 - a_2x - \ddots}}} \quad (28)$$

has Hankel determinants H_n . So multiple formal power series over K realize the same Hankel sequence. What they may lack is the special properties we care about: being quadratic algebraic, satisfying the graded companion law, or being the distinguished Stieltjes choice.

5 Comparison with the construction of Chang and Liu

We have shown that our transported series involves fractional powers of α , at least in the particular form we constructed it. A different quadratic series might conceivably avoid them. Chang and Liu provide a natural comparison, since their construction realizes generic bilateral Somos–4 sequences via Hankel determinants. We now express both constructions in a common notation.

5.1 The shifted Somos–4 orbit

Chang and Liu [1, Sections 3.3–4.2] treat the generic bilateral Somos–4 initial-value problem by Hankel determinants of quadratic orthogonal pairs. Their normalized recurrence is

$$\sigma_{n+2}\sigma_{n-2} = a\sigma_{n+1}\sigma_{n-1} + b\sigma_n^2, \quad n \in \mathbb{Z}, \quad (29)$$

with $\sigma_{-1} = x$, $\sigma_0 = 1$, $\sigma_1 = y$, and $\sigma_2 = z$. Our freely prescribed determinants fit this normalization after the shift

$$\sigma_n = \frac{H_{n+2}}{m}, \quad (x, y, z; a, b) = \left(\frac{\lambda}{m}, \frac{r}{m}, \frac{\eta}{m}; \alpha, \tau \right). \quad (30)$$

This shift uses all six parameters. In particular, it avoids the unwanted constraint $\eta = \alpha\lambda r + \tau m^2$ that would follow from imposing our recurrence at $N = 0$ with $H_0 = 1$.

Write $u^2 = \alpha$ and define

$$\mathcal{N} = \tau r^2 m + \lambda^2 \eta^2 + \alpha \eta m^3 + \alpha \lambda r^3, \quad (31)$$

$$\mathcal{N}_0 = \tau r^2 m - \lambda^2 \eta^2 + \alpha \eta m^3 + \alpha \lambda r^3. \quad (32)$$

Under (30), the Chang–Liu parameters include

$$d_0 = \frac{\lambda r}{m^2}, \quad u_0 = \frac{2r}{m}, \quad (33)$$

$$\frac{v}{u} = \frac{\mathcal{N}}{2\alpha\lambda r\eta m}, \quad uv = \frac{\mathcal{N}}{2\lambda r\eta m}, \quad (34)$$

$$\frac{v_0}{u} = \frac{\mathcal{N}_0}{2\alpha\lambda r\eta m}, \quad uv_0 = \frac{\mathcal{N}_0}{2\lambda r\eta m}, \quad (35)$$

and

$$f = \frac{\tau}{\alpha} - \frac{\mathcal{N}^2}{4\alpha\lambda^2 r^2 \eta^2 m^2}. \quad (36)$$

The sign difference between $\mathcal{N}$ and $\mathcal{N}_0$ is essential because v_0 enters the distinguished moment branch. The scaling $X = uZ$ turns their curve into

$$Y^2 = (\alpha Z^2 + f)^2 + 4\alpha Z - 4uv, \quad (37)$$

which is defined over K . Again, this statement concerns the curve only. It says nothing yet about whether the coefficients of a chosen generating series lie in K .

5.2 Translation to Jacobi data

Use the local Jacobi convention

$$F(t) = \frac{\lambda}{1 - A_0 t - \frac{B_1 t^2}{1 - A_1 t - \frac{B_2 t^2}{1 - A_2 t - \frac{B_3 t^2}{1 - A_3 t - B_4 t^2 M_4(t)}}}}, \quad M_4(0) = 1. \quad (38)$$

The product formula

$$H_N = \lambda^N \prod_{j=1}^{N-1} B_j^{N-j} \quad (39)$$

gives

$$B_1 = \frac{m}{\lambda^2}, \quad B_2 = \frac{\lambda r}{m^2}, \quad B_3 = \frac{m\eta}{r^2}, \quad B_4 = \frac{r(\alpha\eta m + \tau r^2)}{\lambda\eta^2}. \quad (40)$$

The companion identities through $D_3 = H_5$ force

$$A_0 = \frac{r}{\lambda}, \quad A_1 = \frac{\eta\lambda^2 + m^2}{\lambda m r}, \quad A_2 = \frac{\alpha\eta m^3 + \lambda r^3 + \tau m^2 r^2}{\eta\lambda m r}. \quad (41)$$

Proposition 10 (Fourth shifted determinant). *For the convention (38),*

$$D_4 = \lambda^4 B_1^3 B_2^2 B_3 (A_0 A_1 A_2 A_3 - A_0 A_1 B_3 - A_0 A_3 B_2 - A_2 A_3 B_1 + B_1 B_3). \quad (42)$$

After substitution of (40) and (41), this reduces to

$$D_4 = \frac{\eta}{\lambda r^2} \left[A_3 r (\alpha\eta m + \tau r^2) - \eta^2 \lambda \right]. \quad (43)$$

Consequently $D_4 = H_6$, where

$$H_6 = \frac{\alpha^2 \eta m r + \alpha \tau r^3 + \tau \eta^2 \lambda}{\lambda m},$$

uniquely forces

$$A_3 = \frac{\alpha^2 \eta m r^3 + \alpha \tau r^5 + \eta^3 \lambda m + \tau \eta^2 \lambda r^2}{\eta m r (\alpha\eta m + \tau r^2)}. \quad (44)$$

Proof. Equation (42) follows by direct expansion of the shifted 4×4 Hankel determinant in Jacobi parameters. Substitution and collection of the terms linear in A_3 gives (43). Solving (43) for A_3 after inserting the displayed value of H_6 gives (44). $\square$

An equivalent useful form is

$$A_3 = \frac{\alpha r^2}{\eta m} + \frac{\eta \lambda (\eta m + \tau r^2)}{mr(\alpha \eta m + \tau r^2)}.$$

No B_4 occurs in D_4 : the fourth ordinary Hankel step determines B_4 , while the fourth shifted step determines A_3 , without imposing any extra relation among the six parameters.

Chang and Liu use

$$d_n = \frac{\sigma_{n-1}\sigma_{n+1}}{\sigma_n^2}.$$

Substitution of (30) gives the exact all-order interface

$$d_n = \frac{H_{n+1}H_{n+3}}{H_{n+2}^2} = B_{n+2}. \quad (45)$$

Thus $d_0 = B_2$, $d_1 = B_3$, and $d_2 = B_4$. Thus the subdiagonal parameters match after an index shift: only B_1 comes before the Chang–Liu sequence begins. The diagonal parameters and the normalization of the generating-series coefficients still have to be treated separately.

6 The parity involution in the Chang–Liu branch

The same parity symmetry appears in Chang and Liu’s construction, though dressed as a choice of square root. Let

$$S(t; u) = \sum_{n \geq 0} s_n(u) t^n, \quad u^2 = \alpha,$$

be their distinguished moment branch. Chang and Liu show (Remark 4.7):

$$S(t; -u) = S(-t; u), \quad s_n(-u) = (-1)^n s_n(u). \quad (46)$$

The involution $u \mapsto -u$ is precisely the moment-parity involution from Proposition 12. Consequently,

$$H_N(S(t; -u)) = H_N(S(t; u)), \quad D_N(S(t; -u)) = (-1)^N D_N(S(t; u)). \quad (47)$$

So the two sign choices for u do not yield two different ordinary Somos–4 sequences. Instead, they give coefficient sequences related by alternating signs. Ordinary Hankel determinants cannot see this difference, but shifted ones can.

To see this directly in the coefficients, write

$$S(t; u) = E(t^2) + utR(t^2),$$

where E and R have even-indexed moment series. By (46), the even moments are invariant under $u \mapsto -u$ while the odd moments change sign. So the appearance of $u = \sqrt{\alpha}$ in this moment branch records the parity orientation of the odd moments. This observation applies to the particular lift we have chosen; it does not claim that every representation of the ordinary Hankel orbit requires a quadratic extension.

The shifted-determinant normalization makes the choice concrete. If one branch satisfies $D_N = c_N H_{N+2}$, its parity conjugate satisfies $D_N = (-1)^N c_N H_{N+2}$. Picking a literal or graded companion law thus selects a parity orientation. This is all the involution contributes here.

7 Summary

Let us collect the main threads.

Our weighted transport (7) yields a quadratic algebraic series over $K(q)$ (with $q^6 = \alpha$) whose first four ordinary Hankel determinants are (λ, m, r, η) and whose full Hankel orbit satisfies (10).

The once-shifted determinants obey the graded companion law (11). Already the coefficient $b_1 = q^{-5}r$ lies outside K generically, proving that this particular transported branch does not descend.

Yet the curve itself descends: it has the rational model (25) over K . This shows that the orbit and the curve can descend even when the generating series does not.

The dictionary (30) links our orbit to Chang and Liu's construction, with subdiagonal interface $d_n = B_{n+2}$ holding throughout. The involution $u \mapsto -u$ in their moment branch is precisely moment parity: it fixes every ordinary Hankel determinant but multiplies each once-shifted determinant of order N by $(-1)^N$.

8 Conclusion

The construction is elegant and the limitation is real.

On the positive side, the weighted substitution (7) with $q^6 = \alpha$ extends our five-parameter generating function to the full six-parameter Somos-4 recurrence. The first four determinants stay at (λ, m, r, η) , and the whole sequence satisfies (10).

On the other hand, generating-series coefficients carry more information than Hankel determinants. Under the same rescaling, the shifted determinants follow (11), and immediately $b_1 = q^{-5}r$ escapes from $K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$ in the generic case. The Hankel sequence itself lies in K , and the quartic (25) shows the curve does as well. Three seemingly parallel rationality questions have different answers: the determinants and curve are rational in the six parameters, but this particular power series is not.

The Chang-Liu comparison clarifies the remaining ambiguity. Their identity $S(t; -u) = S(-t; u)$ means that flipping the sign of $u = \sqrt{\alpha}$ merely alternates the signs of odd coefficients. This preserves all ordinary Hankel determinants but changes each once-shifted determinant of order N by $(-1)^N$. In short: ordinary Hankel determinants lose a parity choice that the shifted determinants retain. No deeper geometric interpretation is needed.

Disclosure on authoring and verification tools

The manuscript was revised with assistance from AI-based language and symbolic calculation tools. Their use was directed toward notation control, LaTeX editing, and exact algebraic checks. The author remains responsible for the mathematical statements and the final text.

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